

4-Channel Touchless Switchboard For The Post Covid World

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As Covid-19 is still rampant we need to be cautious. Wearing mask, social distancing, and washing or sanitising our hands properly from time to time is absolutely necessary to avoid it. As a precaution, we should also avoid touching any item including the electrical switches as far as possible. In public places, schools, and offices the electrical switchboards are touched by one and all, and so can pass on the virus from one person to many.

Here is a solution in the form of a 4-channel touchless switchboard.

Arduino Uno as the heart of the project, surrounded by four ultrasonic sound sensors, offers a simple, low-cost, and novel solution. The switches can be turned on and off alternately by just placing hand next to the ultrasonic sound sensors on the device, without touching.

Circuit and working

Fig. 1 shows the circuit diagram of 4-channel touchless switchboard. It comprises four HC-SR04 ultrasonic sound sensors (SEN1-SEN4), four-channel relay board (RM1), and

Arduino Uno board (Board1).

The microcontroller in Arduino Uno board receives the switching signals from the sensors and the program code sends the signals to the 4-channel relay module and thereby switches on/off the devices. The ultrasonic sound sensor HC-SR04 is used to sense the switching action.

Pin details of the HC-SR04 ultrasonic sensor are shown in Fig. 2. It has following four pins:

- VCC: +5V DC
- Trig: Trigger (input)
- Echo: Echo (output)
- GND: GND

The sensors can be easily interfaced with Arduino Uno. When a hand is placed about 5cm from the sensor the program in Arduino reads the signal and the relay is energised to perform the switching action. When hand is again placed near the sensor, the Arduino again reads the signal and the relay is de-energised to toggle the switch from its previous position. Similar action is performed with all other ultrasonic sound sensors.

Table 1 shows the comparison between various types of switchboards that are generally available. Table 2 shows the Arduino Uno pin connections with components.

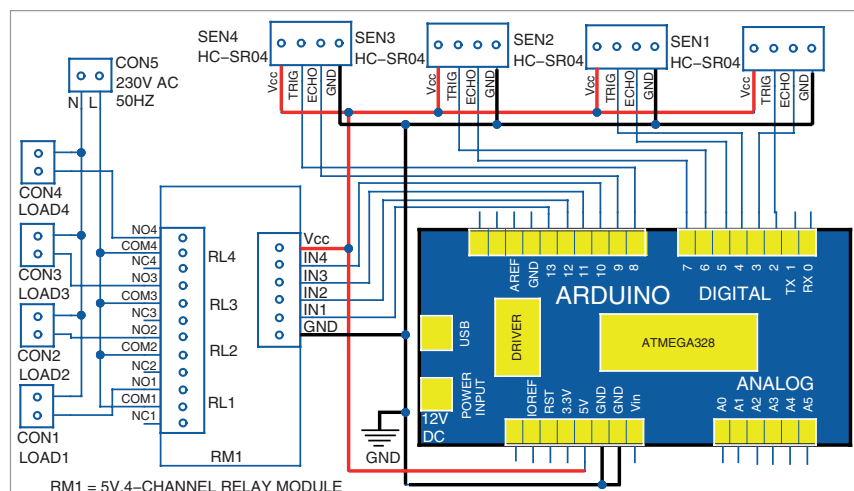


Fig. 1: Circuit diagram of the touchless switchboard

Types of Switchboards		Table 1
Switchboards	Details	
Traditional switchboards	 - Highly rugged - Complete manual operation	
IR based wireless switchboards	 - Operate with IR remote - Appliance control is done by the relay in the background - Relay board operates on 12V DC	
Wireless RF transmitter receiver based remote control switchboard	 - Operates on RF (radio frequency) 433MHz wireless remote - Relay board operates on 12V DC - Appliance control is done by the relay in the background	
Wi-Fi based wireless switchboard	 - Wi-Fi wireless switch works with Alexa, Google Home, Nexa - Mobile application - Wireless standard: 802.11 b/g/n security - Mostly it has internal DC power supply	

4-channel relay board

Fig. 3 shows a 4-channel relay board. This is a low-level, 5V board whose each channel consumes 15-20mA driver current. It can be used to run various appliances that consume large



Fig. 2: Pin details of HC-SR04 sensor